

RL-Driven High-Interaction IoT Honeypot for Adaptive Attacker Engagement

Dr. Neha Agarwal
dept. Computer Science

Indian Institute of Information Technology, Sri City
Andhra Pradesh, India
email

Kainuru Balaji
dept. Computer Science

Indian Institute of Information Technology, Sri City
Andhra Pradesh, India
balaji.k23@iiits.in

Abstract—The rapid proliferation of Internet of Things (IoT) devices across industrial and consumer environments has significantly expanded the attack surface for cyber threats. Conventional IoT security mechanisms are often limited in adaptability and can be easily bypassed by sophisticated attackers. While traditional honeypot systems provide a means to observe adversarial behavior, they typically rely on static configurations and predefined responses, making them susceptible to detection and evasion. This paper proposes an adaptive multi-honeypot framework designed for IoT environments, leveraging Reinforcement Learning (RL) to enhance interaction realism and attacker engagement. The system integrates multiple protocol-specific honeypots and employs a Deep Q-Network (DQN)-based decision engine to dynamically respond to attacker actions. Communication within the environment is facilitated using the MQTT protocol, enabling lightweight and scalable interaction across distributed components. The proposed approach aims to maximize attacker dwell time by generating intelligent, context-aware responses, thereby increasing the volume and quality of captured attack data. This enriched dataset can be utilized to improve threat intelligence, detection strategies, and overall system resilience. Experimental evaluation demonstrates that the adaptive honeypot significantly outperforms traditional static honeypots in terms of stealth, interaction depth, and data collection efficiency.

Index Terms—Internet of Things, Honeypot, Deep Q-Network, MQTT protocol

I. INTRODUCTION

The rapid expansion of Internet of Things (IoT) ecosystems has introduced a massive number of interconnected devices into industrial, healthcare, smart home, and critical infrastructure environments. IoT devices such as embedded routers, smart sensors, industrial gateways, network cameras, and edge controllers commonly expose network-facing services including Secure Shell (SSH), Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), and Telnet for remote administration and data exchange. However, constrained computational resources, insecure default configurations, weak authentication mechanisms, and delayed firmware patching have significantly increased the attack surface of IoT deployments, making them attractive targets for cyber adversaries.

Modern IoT attacks frequently involve automated reconnaissance, credential brute-force attempts, malware propagation, botnet recruitment, privilege escalation, firmware manipulation, and remote command execution. In particular, large-

scale botnets such as Mirai have demonstrated how vulnerable IoT services can be exploited to launch distributed denial-of-service (DDoS) attacks and other coordinated malicious activities. Attackers commonly perform behavioral validation and service fingerprinting before exploitation, attempting to determine whether a target system is genuine or artificially emulated. As a result, traditional low-interaction honeypots with static response logic and predictable behavior are often identified early during the reconnaissance stage, leading to poor attacker retention and shallow interaction depth.

Honeypots are widely deployed as cyber deception systems designed to attract adversaries and capture malicious behavior in controlled environments. Conventional honeypot systems are generally classified into low-interaction and high-interaction architectures. Low-interaction honeypots emulate limited services using predefined responses and lightweight protocol simulation, enabling scalable deployment with minimal risk. However, their limited realism and repetitive outputs reduce their effectiveness against sophisticated attackers. High-interaction honeypots provide richer behavioral visibility by exposing realistic operating environments and command execution capabilities, but they often rely on static interaction strategies and lack adaptive response mechanisms [1], [2], [3].

A critical limitation of existing honeypot systems is the inability to dynamically adapt deception strategies according to attacker behavior. Modern adversaries frequently transition through multiple stages including reconnaissance, file-system exploration, privilege escalation attempts, credential harvesting, malware deployment, and persistence establishment. Static response systems fail to sustain engagement during these evolving attack phases because they expose repetitive interaction patterns, unrealistic privilege behavior, and shallow operational realism. Consequently, attackers often disengage early, limiting the collection of meaningful behavioral intelligence.

To address these limitations, recent research has explored intelligent-interaction honeypots that incorporate reinforcement learning (RL) and adaptive deception mechanisms [4], [5]. Reinforcement learning enables an agent to continuously optimize interaction strategies through sequential observation of attacker behavior and reward-driven policy adaptation. In

cybersecurity environments, RL-based systems can dynamically select deception actions, adjust interaction intensity, and personalize responses according to attacker engagement patterns. However, many existing approaches focus primarily on isolated single-protocol environments and do not support coordinated adaptive deception across heterogeneous IoT services.

Although existing intelligent-interaction honeypots demonstrate improved adaptability compared to static deception systems, several limitations remain. Many prior approaches primarily focus on isolated single-protocol environments such as SSH-only interaction or firmware-level emulation, limiting their ability to capture heterogeneous attacker behavior across multiple IoT services. Furthermore, existing systems often lack coordinated adaptive orchestration mechanisms capable of dynamically adjusting deception intensity, privilege behavior, and information disclosure strategies according to attacker engagement patterns. These limitations reduce interaction realism and restrict the collection of high-quality behavioral intelligence in complex IoT threat environments.

In this work, we propose a fully autonomous RL-driven high-interaction IoT honeypot framework that performs adaptive deception orchestration across multiple protocol services including SSH, HTTP, FTP, and Telnet. The proposed framework models attacker interaction as a sequential decision-making process in which protocol-specific behavioral features are transformed into structured state representations consumed by a Deep Q-Network (DQN)-based decision engine. The system dynamically selects deception strategies including adaptive shell behavior, staged honeypot exposure, controlled privilege escalation responses, delayed interaction (tarpting), malware bait delivery, and deception-oriented failure simulation based on observed attacker activity.

Unlike traditional static honeypots, the proposed framework employs progressive trust calibration to increase attacker investment and sustain long-duration interaction. For example, login attempts may initially experience intentional rejection before controlled access is gradually granted to simulate realistic authentication behavior. Similarly, operational artifacts such as BusyBox-based filesystems, AWS-format credentials, MQTT configuration files, firmware images, database credentials, and SCADA management interfaces are progressively exposed during deeper engagement stages to encourage continued exploration and privilege escalation attempts. This staged information disclosure strategy improves interaction realism while enabling the collection of richer behavioral intelligence including command sequences, scanning patterns, privilege escalation behavior, malware download attempts, and credential harvesting activity.

The framework utilizes a distributed MQTT-based publish-subscribe architecture to coordinate communication between protocol-specific traps and the centralized reinforcement learning engine. Each honeypot service continuously extracts behavioral signals including command frequency, interaction depth, reconnaissance intensity, privilege escalation attempts, and contextual protocol activity to construct state vectors

for adaptive decision-making. Sessions exhibiting aggressive scanner-like behavior are selectively redirected toward tarpit-oriented responses to conserve resources while maintaining deceptive interaction continuity. Furthermore, the DQN agent continues learning after deployment through replay-buffer updates derived from logged attacker sessions stored in MongoDB, enabling continual adaptation to evolving attack strategies.

The primary contributions of this work are summarized as follows:

- We propose a fully autonomous reinforcement learning-driven adaptive cyber deception framework for multi-service IoT environments.
- We design a protocol-aware adaptive deception architecture supporting coordinated interaction across SSH, HTTP, FTP, and Telnet services through centralized RL orchestration.
- We introduce progressive trust calibration and staged honeypot exposure strategies to improve attacker retention, interaction realism, and behavioral intelligence collection.
- We implement a DQN-based adaptive decision engine capable of dynamically selecting deception actions based on protocol-specific behavioral state representations.
- We develop a distributed MQTT-based communication framework enabling scalable low-latency interaction between protocol traps and the RL decision engine.
- We introduce a behavioral evaluation framework consisting of the Behavioral Engagement Index (BEI), Deception Retention Rate (DRR), and Escalation Persistence Score (EPS) to quantitatively measure attacker engagement depth, deception persistence, and privilege-oriented behavioral continuity within adaptive cyber deception systems.

II. RELATED WORK

A. Low-Interaction and High-Interaction Honeypots

Traditional honeypots are generally categorized into low-interaction and high-interaction architectures depending on the level of attacker engagement and system realism provided. Low-interaction honeypots emulate limited protocol behavior using predefined responses and lightweight service simulation, enabling scalable deployment with minimal operational risk. Honeyd introduced one of the earliest virtual honeypot frameworks capable of simulating multiple network services and operating system fingerprints for large-scale attack observation [6]. Similarly, IoT-focused frameworks such as U-PoT and ThingPot extended lightweight deception techniques to IoT-specific environments by emulating protocols including UPnP and IoT messaging services [1], [2]. These systems effectively capture automated scanning and reconnaissance activity; however, their reliance on static response templates and limited interaction realism reduces their ability to sustain long-duration attacker engagement.

High-interaction honeypots provide richer behavioral visibility by exposing realistic operating environments and interactive services to attackers. Systems such as SIPHON deploy

large-scale distributed infrastructures capable of exposing genuine IoT services and collecting malware samples, command traces, and post-compromise activity [7]. Although these systems significantly improve interaction realism and behavioral intelligence collection, they often introduce increased deployment complexity, scalability challenges, and operational risks associated with exposing near-real services. More importantly, both low-interaction and traditional high-interaction honeypots generally rely on static interaction strategies that lack adaptive behavioral response mechanisms. As a result, attackers can identify repetitive interaction patterns during reconnaissance and disengage before deeper behavioral intelligence can be collected.

B. Intelligent-Interaction and Reinforcement Learning-Based Honeypots

To overcome the limitations of static deception systems, recent research has increasingly focused on intelligent-interaction honeypots that integrate adaptive behavioral modeling and reinforcement learning mechanisms. IoTcandyJar introduced one of the early intelligent-interaction IoT honeypots by modeling attacker interaction as a Markov Decision Process (MDP) and applying Q-learning to dynamically select deception responses according to attacker behavior patterns [4]. The framework demonstrated that adaptive response selection can improve attacker engagement compared to conventional static honeypots. However, the approach remains constrained by limited scalability and relatively slow convergence during long-term interaction learning.

FirmPot further enhanced interaction realism by leveraging firmware emulation and sequence-to-sequence response generation to simulate realistic IoT device behavior [5]. By executing actual firmware images, the framework improves protocol fidelity and behavioral authenticity during attacker interaction. Nevertheless, the system primarily focuses on single-device emulation and does not address coordinated adaptive deception across heterogeneous multi-service environments commonly observed in modern IoT infrastructures.

More recently, AIIPot introduced a reinforcement learning-driven intelligent-interaction architecture that combines Bidirectional Encoder Representations from Transformers (BERT)-based request analysis with MDP-based adaptive interaction policies [8]. The framework applies Q-learning to optimize response selection and improve attacker engagement duration through context-aware interaction. Among existing adaptive honeypot systems, AIIPot exhibits conceptual similarities to the proposed framework because both approaches employ reinforcement learning to dynamically guide deception behavior. However, AIIPot primarily targets isolated interaction environments and does not address coordinated multi-service orchestration, protocol-specialized behavioral policies, progressive deception escalation, staged honeypot exposure, or continuous online adaptation across heterogeneous IoT protocols.

Recent learning-based IoT deception research has further emphasized the importance of adaptive behavioral realism

for sustaining long-duration attacker engagement. HoneyIoT models attacker interaction as an MDP and applies reinforcement learning to dynamically generate responses that maximize attacker retention and malware collection efficiency [3]. The framework demonstrates that adaptive interaction policies significantly improve deception effectiveness compared to fixed-response honeypots by learning attacker-specific behavioral patterns from real-world attack traces. However, existing learning-based approaches still primarily focus on interaction optimization within isolated service environments and provide limited support for coordinated protocol-aware adaptive deception across distributed IoT attack surfaces.

Although intelligent-interaction honeypots significantly improve deception realism and attacker engagement compared to traditional static systems, several important limitations remain unresolved. Most existing frameworks concentrate on isolated service environments, lack coordinated adaptive orchestration across heterogeneous protocols, and provide limited support for progressive deception strategies capable of dynamically evolving according to attacker behavior. In contrast, the proposed framework introduces a unified adaptive deception architecture that combines protocol-specialized reinforcement learning agents, progressive trust calibration, staged honeypot exposure, adaptive tarpit mechanisms, realistic operational IoT simulation, and continuous online learning across SSH, HTTP, FTP, and Telnet services to improve attacker engagement depth and behavioral intelligence collection in heterogeneous IoT environments.

C. Adaptive and Dynamic Reinforcement Learning Approaches for Cyber Deception

Recent research has explored reinforcement learning techniques for improving adaptability in dynamic cyber deception environments where attacker behavior continuously evolves over time. Unlike conventional static honeypots that rely on predefined responses and fixed deployment strategies, adaptive reinforcement learning-based approaches aim to optimize deception policies through continuous interaction feedback and behavioral learning.

Learning-based IoT honeypot systems have demonstrated the importance of modeling attacker interaction as a sequential decision-making process capable of dynamically adjusting responses according to attacker behavior [3]. HoneyIoT leverages reinforcement learning to model attacker interaction using a Markov Decision Process (MDP) and dynamically selects responses that maximize attacker engagement and malware collection efficiency. By analyzing real-world attack traces collected from deployed IoT devices, the framework improves deception realism and sustains attacker interaction more effectively than fixed-response honeypots. The study further highlights the significance of adaptive response generation for bypassing reconnaissance behavior and maintaining long-duration attacker engagement. However, the framework primarily focuses on interaction optimization within individual IoT device environments and does not address coordinated

multi-service adaptive deception or protocol-specialized orchestration across heterogeneous IoT infrastructures.

Recent reinforcement learning-driven cyber deception research has also investigated adaptive honeypot deployment strategies for improving resource utilization and attack-surface coverage in dynamic environments. Reinforcement learning-based deployment frameworks model attacker-defender interaction using Markov Decision Processes in which honeypot services and deployment locations are continuously adjusted according to observed attacker behavior patterns [9]. These approaches demonstrate that adaptive deployment policies can significantly improve deception efficiency compared to static placement mechanisms while reducing unnecessary resource consumption. Nevertheless, such systems primarily emphasize deployment optimization and dynamic resource allocation rather than sustained high-interaction attacker engagement, realistic behavioral simulation, or protocol-aware deception adaptation.

In parallel, lightweight IoT honeynet architectures have explored dynamic honeypot management techniques for improving scalability and operational efficiency in constrained IoT environments [10]. MQTT-based lightweight honeynet systems dynamically adjust the number of active honeypot services according to observed attack frequency and network activity, enabling more efficient resource utilization across IoT infrastructures. Although these approaches improve deployment flexibility and scalability, they generally rely on static interaction logic and lack intelligent adaptive response generation capable of evolving according to attacker behavior.

Overall, existing adaptive and reinforcement learning-based cyber deception systems demonstrate substantial progress toward intelligent honeypot architectures; however, several important limitations remain unresolved. Many current approaches primarily focus on isolated protocol environments, deployment optimization, or interaction adaptation within individual services while providing limited support for coordinated multi-service orchestration, progressive deception escalation, protocol-specialized adaptive policies, and continuous online learning in heterogeneous IoT attack environments. In contrast, the proposed framework integrates protocol-specialized reinforcement learning agents, adaptive behavioral state modeling, progressive trust calibration, staged honeypot exposure, adaptive tarpit strategies, realistic operational IoT simulation, and continuous online learning within a unified multi-service deception architecture designed specifically for heterogeneous IoT infrastructures.

III. SYSTEM ARCHITECTURE AND DESIGN

A. System Overview

The proposed framework is designed as a distributed adaptive cyber deception architecture for heterogeneous IoT environments. The system integrates multiple protocol-specialized honeypot services with reinforcement learning-driven adaptive response generation to sustain realistic attacker interaction and maximize behavioral intelligence collection. Unlike conventional static honeypots that rely on predefined interaction tem-

plates, the proposed architecture continuously adapts deception strategies according to attacker behavior across multiple IoT-facing services including SSH, HTTP, FTP, and Telnet.

The framework is deployed within an isolated Ubuntu 22.04 virtualized environment using containerized service orchestration to ensure controlled experimentation and safe attacker interaction. The architecture consists of four protocol-specific deception services, independent Deep Q-Network (DQN)-based adaptive agents, a lightweight MQTT-based communication layer, and a centralized MongoDB logging infrastructure. Each protocol service maintains its own reinforcement learning policy network, replay buffer, target network, and adaptive interaction strategy to preserve protocol-aware behavioral specialization.

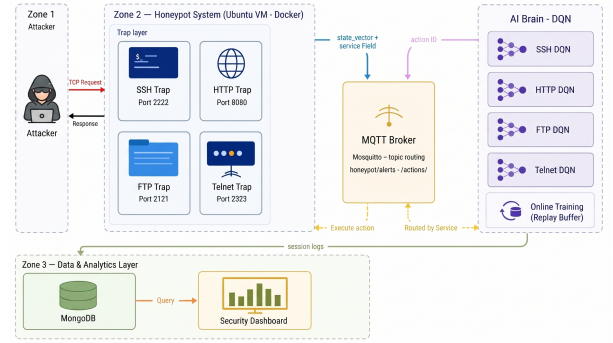


Fig. 1: System Architecture of the Adaptive Deception Framework.

Fig. 1 illustrates the overall adaptive deception pipeline of the proposed framework. When an attacker interacts with a protocol service, the corresponding trap extracts behavioral features including command frequency, interaction depth, privilege escalation attempts, and threat-level indicators. These behavioral observations are transformed into structured state vectors and transmitted asynchronously through MQTT publish-subscribe messaging. The service-specific DQN agent subscribes to the corresponding interaction topic, performs adaptive policy inference, and returns an optimal deception action to the originating protocol trap.

The selected deception strategy is then executed dynamically within the interaction environment. Depending on the observed attacker behavior, the framework may perform progressive trust escalation, staged honeypot exposure, adaptive shell interaction, delayed-response tarpitting, fake privilege escalation handling, malware bait delivery, or deception-oriented controlled failures. This event-driven interaction model enables low-latency adaptive response generation while maintaining loose coupling between distributed components.

Unlike traditional static honeypot architectures, the proposed framework emphasizes protocol-specialized adaptive deception. Each service implements a distinct interaction model optimized for its corresponding attack surface. SSH and Telnet services prioritize command-line realism and privilege escala-

tion engagement, HTTP services focus on SCADA-oriented credential harvesting and firmware upload interaction, while FTP services emphasize malware capture and firmware repository exploration. This specialization enables the framework to capture heterogeneous attacker behaviors across multiple operational IoT attack surfaces.

Furthermore, the architecture incorporates progressive trust calibration to improve attacker retention and interaction depth. Rather than immediately granting unrestricted access, selected services intentionally introduce controlled authentication failures and gradual interaction escalation to simulate realistic operational environments. Honeytokens, fake credentials, firmware artifacts, MQTT configurations, and cloud-related operational files are strategically exposed only after sufficient interaction depth has been established. This staged deception strategy increases operational realism while encouraging continued attacker exploration and privilege escalation attempts.

The framework operates as a fully autonomous adaptive deception system in which response decisions are selected dynamically by service-specific reinforcement learning agents without requiring manual intervention during active sessions. During periods of inactivity, protocol services remain event-driven and idle while awaiting MQTT-triggered attacker interaction events, reducing unnecessary computational overhead. All attacker interactions, state transitions, selected actions, command sequences, malware download attempts, and behavioral observations are continuously logged within MongoDB for post-analysis and online reinforcement learning updates.

The architecture additionally incorporates adaptive scanner-aware interaction handling. Sessions exhibiting aggressive reconnaissance behavior, repetitive automated command execution, or high-frequency probing activity are dynamically redirected toward tarpit-oriented deception strategies involving delayed command execution and controlled interaction slowdown. This selective resource-aware deception mechanism enables the framework to conserve computational resources while maintaining realistic interaction continuity against automated scanning activity.

Overall, the proposed system establishes a scalable and adaptive cyber deception framework capable of performing protocol-aware behavioral engagement, continuous online learning, and distributed adaptive response orchestration across heterogeneous IoT services.

B. Protocol-Specialized Trap Service Design

The proposed framework incorporates protocol-specialized deception environments designed to emulate realistic IoT operational behavior while maximizing attacker engagement depth and behavioral intelligence collection. Unlike conventional honeypots that rely on static service emulation, each protocol service implements adaptive deception strategies optimized for distinct attacker objectives, interaction patterns, and reconnaissance behaviors. The framework emphasizes operational realism, progressive trust escalation, staged information disclosure, and protocol-aware behavioral engagement to reduce

honeypot fingerprinting and sustain long-duration attacker interaction.

A key design principle of the framework is progressive deception escalation. Rather than immediately exposing high-value assets or unrestricted system access, the services gradually increase interaction realism and operational visibility according to attacker behavior. Initial interaction stages primarily expose lightweight operational artifacts such as banners, directory structures, and system information. As engagement depth increases, the framework progressively reveals configuration files, network details, shell histories, fake credentials, API keys, privilege escalation opportunities, and high-value honeytokens. This staged disclosure strategy mimics the natural discovery process observed in real post-compromise environments and increases attacker trust while reducing early disengagement.

1) *SSH Deception Environment*: The SSH deception service operates on port 2222 and represents the primary high-interaction environment within the framework. The service is implemented using the Paramiko library and emulates a BusyBox-oriented embedded Linux environment commonly observed in operational IoT devices. BusyBox was intentionally selected because many real-world IoT systems utilize lightweight embedded Linux distributions that expose minimal command-line environments and simplified filesystem structures.

Unlike conventional SSH honeypots that provide limited command simulation, the proposed SSH environment supports controlled interactive command execution combined with adaptive deception strategies. The service intentionally introduces one or more authentication failures before granting access in order to simulate realistic operational behavior and establish progressive trust calibration. This mechanism increases attacker investment and reduces suspicion by mimicking natural authentication inconsistencies frequently observed in real systems.

After authentication, attackers enter a high-realism shell environment containing operational artifacts including `/etc/shadow`, cloud credential files, MQTT configurations, AWS-format API keys, shell histories, configuration logs, and simulated administrative resources. Among these artifacts, shell history files play a particularly important role in maintaining operational consistency because they expose believable administrator activity traces such as backup execution, configuration edits, cloud synchronization commands, and service management operations. This continuity of operational evidence significantly increases environmental realism and reinforces attacker trust.

The SSH service dynamically adapts interaction behavior using a protocol-specialized DQN agent capable of selecting multiple deception strategies including limited shell access, unrestricted interactive shells, fake privilege escalation handling, staged honeypot exposure, adaptive tarpitting, and deception-oriented controlled failures. The fake sudo mechanism repeatedly requests authentication credentials and selectively grants simulated privilege escalation after sufficient

interaction depth has been established. This adaptive privilege handling encourages attackers to continue exploring the environment while enabling the framework to observe privilege escalation behavior and post-compromise interaction patterns.

The SSH environment consistently demonstrated the deepest attacker engagement during experimentation because attackers psychologically perceived the system as genuinely compromised once interactive shell access was obtained. The combination of filesystem realism, unrestricted exploration capability, operational artifacts, privilege escalation opportunities, and staged credential exposure transformed attacker behavior from simple reconnaissance into extended exploratory interaction. Unlike request-oriented protocols such as HTTP or FTP, the command-driven SSH environment enabled continuous feedback-oriented interaction loops that significantly increased dwell time and interaction depth.

2) *HTTP Deception Environment*: The HTTP deception service operates on port 5000 and emulates a SCADA-oriented IoT gateway administrative interface designed to encourage credential harvesting, firmware upload attempts, and administrative exploration behavior. The environment presents a dynamically updating telemetry dashboard that mimics operational industrial monitoring systems frequently observed in IoT and industrial control deployments.

To improve realism, the service embeds MQTT credentials, database passwords, API keys, and administrative artifacts within HTML comments and hidden interface components. These operational artifacts are intentionally exposed progressively during interaction to encourage deeper administrative exploration and credential-oriented behavior. Firmware upload functionality is additionally implemented to capture uploaded payloads and malicious files submitted through HTTP POST requests.

Compared to traditional web-oriented honeypots that primarily capture scanning activity, the proposed HTTP environment emphasizes deception-oriented operational realism through dashboard interaction, hidden credential exposure, and simulated industrial management functionality.

3) *FTP Deception Environment*: The FTP deception service operates on port 2121 and emulates a realistic IoT firmware repository using a vsFTPD 3.0.3-oriented interaction environment. The service supports both active and passive transfer modes and exposes operational firmware storage directories designed to simulate production IoT device maintenance infrastructure.

The environment contains multiple deception-oriented artifacts including firmware images, MQTT configuration files, cloud credential archives, API key repositories, and device configuration files. Uploaded payloads and transferred malware artifacts are continuously logged within MongoDB for forensic analysis and behavioral intelligence collection. The FTP environment is specifically optimized to encourage malware deployment activity, firmware exploration, and credential-oriented filesystem traversal behavior.

4) *Telnet Deception Environment*: The Telnet deception service operates on port 2323 and emulates a BusyBox-based

IoT router shell environment optimized for botnet attraction and malware propagation analysis. The service intentionally mimics operational embedded router behavior using realistic banners, network configuration outputs, embedded device artifacts, and lightweight command-line interaction.

A specialized deception mechanism referred to as the *Mirai Magnet* is implemented to attract Mirai-like malware propagation behavior and automated payload deployment attempts. Commands such as `wget` and `curl` appear operationally successful in order to encourage malware delivery behavior while simultaneously capturing payload URLs and attacker interaction patterns for forensic analysis. The environment additionally exposes simulated NVRAM dumps and operational router credentials to improve realism and sustain attacker exploration.

The Telnet service further incorporates adaptive tarpitting strategies involving randomized command execution delays ranging from 5 to 8 seconds. This delayed-response mechanism simulates constrained IoT hardware performance while selectively slowing aggressive automated scanning activity. By dynamically increasing interaction latency during scanner-oriented behavior, the framework conserves computational resources while maintaining believable operational realism.

Overall, the protocol-specialized trap environments collectively establish a distributed adaptive deception ecosystem capable of sustaining realistic attacker interaction across heterogeneous IoT attack surfaces while maximizing behavioral intelligence collection and operational deception realism.

C. Behavioral State Representation

The proposed framework models attacker interaction as a sequential behavioral decision process in which observable attacker activity is continuously transformed into protocol-aware state representations for adaptive deception optimization. Rather than relying solely on static protocol events, the framework abstracts attacker behavior into structured behavioral indicators that capture interaction intensity, exploration depth, privilege escalation intent, and malware-oriented activity.

To ensure compatibility across heterogeneous IoT services, the framework represents attacker interaction using a fixed-dimensional behavioral state vector defined as:

$$S = [f_1, f_2, f_3, f_4, f_5] \quad (1)$$

where each feature corresponds to a specific behavioral characteristic extracted from active attacker interaction. The state representation is intentionally designed to maintain protocol-independent consistency while preserving protocol-specific behavioral semantics.

The feature f_1 represents interaction intensity and captures the scale of attacker activity such as command length, payload size, or request complexity depending on the active protocol. Feature f_2 models interaction frequency through command counts, request rates, login attempts, or repeated interaction events. Feature f_3 corresponds to dynamically estimated threat

intensity, while f_4 captures engagement depth associated with attacker exploration progression. Finally, feature f_5 represents protocol-specific contextual behavior including privilege escalation attempts, scanner indicators, file transfer activity, or administrative interaction patterns.

Threat intensity is dynamically estimated using command-semantic indicators associated with malware deployment, payload retrieval, persistence establishment, and destructive activity. Commands including `wget`, `curl`, `chmod`, `python`, `perl`, `bash`, `sh`, `nc`, `dd`, `tftp`, and `ftpget` are treated as high-risk behavioral indicators because they frequently correspond to malware propagation, payload execution, privilege escalation, or persistence-oriented attacker activity. Sessions exhibiting these behaviors receive elevated threat scores within the behavioral state representation.

Engagement depth is modeled using interaction progression metrics including command frequency, filesystem traversal behavior, privilege escalation attempts, session duration, and exploration continuity. Attackers demonstrating repeated exploratory interaction, deeper filesystem navigation, or prolonged command-driven engagement are assigned higher engagement-depth values. This representation enables the framework to distinguish between shallow reconnaissance behavior and sustained post-compromise exploration activity.

Unlike traditional intrusion detection systems that prioritize classification-oriented feature extraction, the proposed framework emphasizes behavioral interaction modeling for adaptive deception optimization. The state representation is therefore designed not only to identify malicious behavior but also to estimate attacker intent, interaction persistence, and deception susceptibility.

TABLE I: Protocol-Specific Behavioral State Feature Mapping Here, II: Input Intensity, IC: Interaction Count, TL: Threat Level, ED: Engagement Depth, CF: Context Feature.

Service	II (f_1)	IC (f_2)	TL (f_3)	ED (f_4)	CF (f_5)
SSH	Command Length	Command Count	1–5	Eng. Depth	Sudo Attempts
HTTP	Payload Size	Request Count	1–5	Eng. Depth	Scanner Ind.
FTP	Command Length	Login Att.	1–5	Eng. Depth	File Trans.
Telnet	Command Length	Command Count	1–5	Eng. Depth	Login Att.

Table I illustrates how behavioral features are specialized across heterogeneous IoT services while maintaining a unified reinforcement learning representation. Although the overall state structure remains fixed-dimensional, each protocol interprets interaction semantics differently according to its operational behavior and attack surface characteristics. This design enables protocol-aware adaptive deception while preserving consistent DQN-based decision modeling across distributed services.

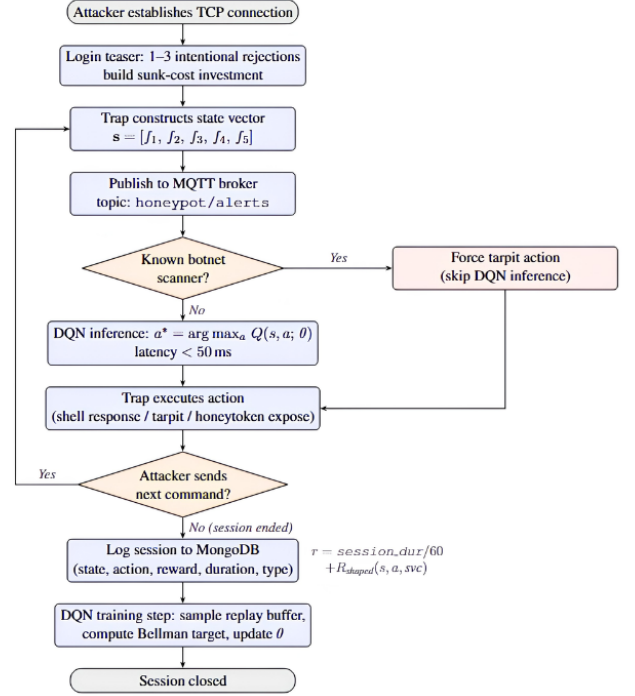


Fig. 2: Operational workflow of the proposed reinforcement learning-driven adaptive deception framework.

D. Deep Reinforcement Learning-Based Adaptive Decision Engine

The adaptive deception mechanism is implemented using protocol-specialized Deep Q-Network (DQN) agents that continuously optimize deception strategies according to evolving attacker behavior. Each protocol service maintains an independent DQN policy network, target network, replay buffer, and exploration policy to preserve protocol-aware adaptive specialization across heterogeneous IoT services.

The attacker interaction environment is modeled as a Markov Decision Process (MDP), where the current behavioral state (s) represents observed attacker activity and the selected action (a) corresponds to an adaptive deception strategy executed within the active honeypot environment. Following each interaction event or attacker state transition, the corresponding protocol service generates an updated behavioral state vector and asynchronously submits it to the associated DQN agent through MQTT-based messaging.

The DQN agent dynamically selects deception actions including unrestricted shell access, staged privilege escalation handling, adaptive tarpitting, controlled interaction failure, staged honeytoken exposure, firmware bait interaction, and malware-oriented deception responses. Action selection is performed using an epsilon-greedy exploration strategy in which the framework initially prioritizes behavioral exploration and gradually shifts toward exploitation of learned deception policies as training progresses. The exploration parameter decays progressively over time, enabling the framework to balance adaptive experimentation and policy convergence.

The DQN architecture is implemented using the PyTorch framework and consists of fully connected neural layers optimized for low-dimensional behavioral state inference. The policy network approximates the action-value function ($Q(s,a)$), enabling the agent to estimate the long-term behavioral utility of each deception strategy under varying attacker interaction conditions.

To stabilize training, the framework employs experience replay buffers containing 10,000 interaction experiences represented as tuples of the form $((s,a,r,s',d))$. Replay-based training breaks temporal correlations between sequential attacker interactions and significantly improves learning stability and convergence performance. Mini-batch experiences are sampled randomly from the replay memory during optimization using the Adam optimizer and Mean Squared Error (MSE) loss function.

The DQN optimization objective follows the Bellman equation:

$$L(\theta) = \mathbb{E} \left[\left(r + \gamma \max_{a'} Q_{\theta-}(s', a') - Q_{\theta}(s, a) \right)^2 \right] \quad (2)$$

where r denotes the observed reward, γ is the discount factor, $Q_{\theta}(s, a)$ represents the policy network, and $Q_{\theta-}(s', a')$ denotes the target network used for stable Q-value estimation. The objective of the DQN optimization process is to minimize temporal-difference error while improving long-term adaptive deception policy selection.

Initial DQN bootstrapping was performed using protocol-filtered NSL-KDD interaction records to stabilize adaptive policy initialization prior to live deployment. Approximately 44,756 unique NSL-KDD records were utilized across SSH, HTTP, FTP, and Telnet behavioral categories, generating nearly 168,000 sampled interaction experiences over 12,000 cumulative bootstrap training episodes. Each protocol-specialized DQN agent maintained an independent replay buffer with a capacity of 10,000 experiences and utilized Adam optimization with a learning rate of 0.001. The target network was synchronized every 200 training iterations while ϵ -greedy exploration decayed from 1.0 to 0.05 during adaptive policy convergence. The DQN architecture consisted of fully connected layers following a $5 \rightarrow 64 \rightarrow 64 \rightarrow N$ structure with ReLU activation and Mean Squared Error (MSE) temporal-difference optimization.

To evaluate reinforcement learning stability during offline initialization, temporal-difference training loss convergence was analyzed across protocol-specialized DQN agents during NSL-KDD bootstrap optimization.

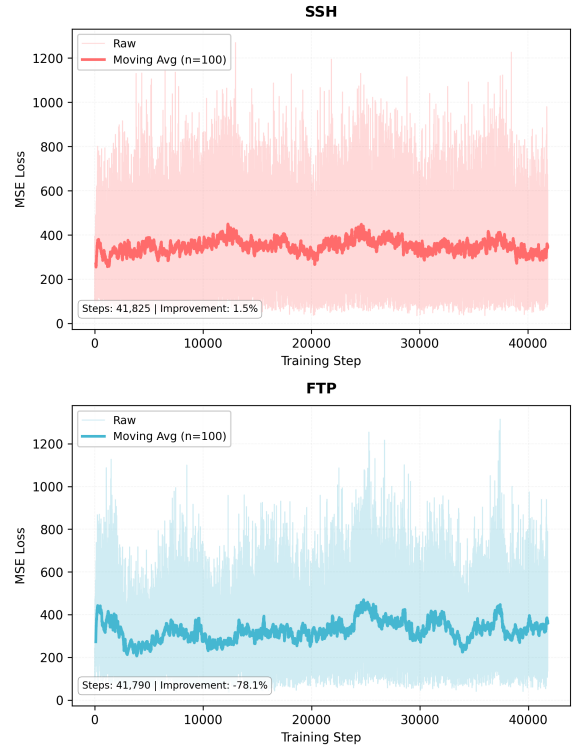


Fig. 3: DQN training loss convergence for SSH and HTTP.

TABLE II: DQN Training Configuration

Parameter	Value
Replay Buffer Size	10,000
Batch Size	64
Learning Rate	0.001
Discount Factor (γ)	0.95
Target Sync Interval	200 steps
Exploration Decay	$1.0 \rightarrow 0.05$
Episodes per Service	$\sim 3,000$
Optimizer	Adam
Loss Function	MSE

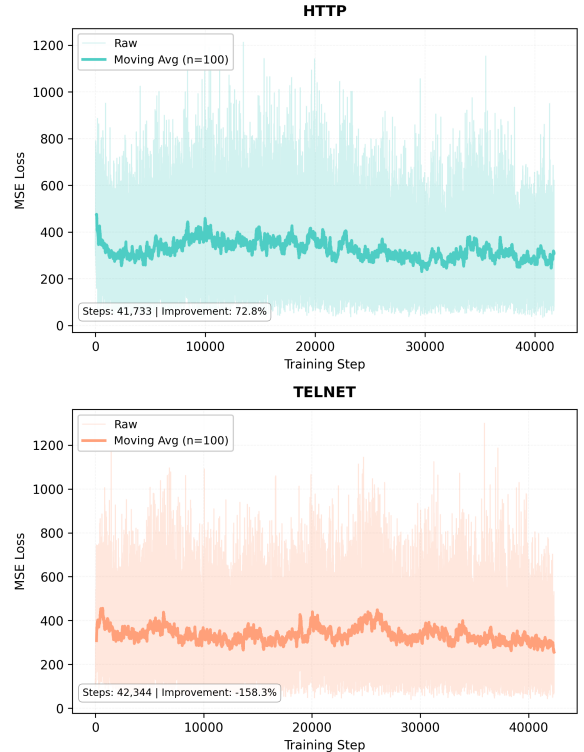


Fig. 4: DQN training loss convergence for FTP and TELNET.

Fig. 3 and Fig. 4 illustrate DQN training convergence behavior observed during offline bootstrap initialization using protocol-filtered NSL-KDD interaction data. Early training stages exhibited unstable temporal-difference loss fluctuations due to exploration-oriented adaptive action selection and randomized replay sampling. As training progressed, moving-average loss curves gradually stabilized across protocol-specialized agents, indicating convergence toward more consistent deception-oriented interaction policies capable of sustaining adaptive attacker engagement behavior. HTTP-oriented interaction policies demonstrated the strongest convergence characteristics due to the larger volume of protocol-specific training samples available during initialization.

The framework employs a hybrid training strategy combining offline initialization with continuous online adaptation. During initial deployment, synthetic experiences derived from the NSL-KDD dataset are used to bootstrap baseline attacker interaction policies and reduce unstable random exploration. After deployment, real attacker interactions continuously generate new behavioral experiences that are logged within MongoDB and incorporated into replay-based online learning updates. This enables the framework to adapt dynamically to evolving attacker strategies, changing reconnaissance patterns, and previously unseen behavioral interactions.

Although SSH and HTTP agents demonstrated relatively stable temporal-difference convergence behavior during offline initialization, the FTP and Telnet agents exhibited comparatively higher loss variance and partial instability during later training stages. This behavior primarily resulted from the smaller volume of protocol-specific NSL-KDD interaction records and increased class imbalance within FTP- and Telnet-oriented behavioral subsets, which produced higher reward variance during replay-based optimization. Despite these fluctuations, the adaptive policies remained operationally stable during live deployment and continued to improve attacker engagement behavior through subsequent online learning updates using real interaction data.

E. Reward Function Design

The reinforcement learning objective of the proposed framework is not merely attack detection, but sustained attacker engagement and behavioral intelligence maximization. Consequently, the reward function is specifically designed to optimize interaction persistence, exploration depth, privilege escalation behavior, and malware-oriented interaction activity.

The reward mechanism combines terminal session rewards with intermediate behavioral shaping incentives:

$$R = \begin{cases} \frac{T}{60}, & \text{if terminal state} \\ \alpha + \beta \cdot f(s, a), & \text{otherwise} \end{cases} \quad (3)$$

where T represents total attacker session duration, while $f(s, a)$ corresponds to context-aware behavioral shaping rewards associated with attacker interaction depth, privilege escalation attempts, filesystem exploration, and malware-oriented activity. The weighting parameters α and β regulate

the contribution of intermediate behavioral rewards during adaptive policy optimization.

Highly rewarding attacker behaviors include sudo-oriented privilege escalation attempts, deep filesystem exploration, credential-oriented traversal, malware upload activity, and repeated command-driven interaction. These behaviors indicate strong attacker engagement and provide valuable post-compromise behavioral intelligence. Conversely, rapid session termination, shallow reconnaissance-only interaction, and repetitive automated scanning activity receive comparatively lower rewards.

A particularly important aspect of the reward mechanism is adaptive deception timing. The reinforcement learning agent continuously learns when to expose credentials, escalate interaction realism, trigger delayed responses, or reveal higher-value operational artifacts in order to maximize attacker persistence. This enables the framework to optimize deception progression dynamically according to observed attacker behavior rather than relying on static interaction templates.

Overall, the reinforcement learning subsystem transforms the honeypot environment into a continuously adaptive behavioral cyber deception system capable of dynamically optimizing attacker interaction realism, engagement depth, and intelligence collection efficiency across heterogeneous IoT attack surfaces.

F. Training Methodology

The training process follows a hybrid approach combining offline initialization with online RL to achieve both stability and adaptability.

In the first phase, the model is bootstrapped using the NSL-KDD dataset. The dataset is not directly used as input to the neural network; instead, it is transformed into synthetic experience tuples of the form (s, a, r, s', d) that align with the defined state representation. These experiences are used to pre-train the DQN agents, enabling them to learn baseline attack patterns and avoid purely random exploration during early deployment.

In the second phase, the system transitions to online learning, where real-time attacker interactions are continuously collected from deployed honeypot services. The observed state transitions and rewards are stored in the replay buffer and used to further refine the policy through stochastic gradient updates. This allows the agent to adapt to evolving attacker behavior and previously unseen attack strategies.

By combining offline bootstrapping with online adaptation, the proposed system achieves faster convergence and improved robustness in dynamic IoT threat environments.

G. MQTT Communication Protocol

The system employs the Message Queuing Telemetry Transport (MQTT) protocol as the communication backbone to enable real-time interaction between distributed components. MQTT is a lightweight publish-subscribe messaging protocol designed for resource-constrained environments and widely used in IoT systems [10]. Its decoupled architecture allows

In the proposed design, honeypot services publish attacker interaction data, including state vectors and event details, to the topic `honeypot/alerts`. The RL decision engine subscribes to this topic, processes the incoming state information, and computes an appropriate response action. The selected action is then published to a service-specific topic of the form `honeypot/actions/<ip>`, where `<ip>` uniquely identifies the attacker session. Each honeypot service subscribes to its corresponding action topic and executes the received instruction.

IV. EXPERIMENTAL EVALUATION AND ANALYSIS

The honeypot framework was evaluated within a controlled local virtualized environment using Docker-based service isolation on an Ubuntu 22.04 infrastructure. The evaluation incorporated structured attacker simulation scenarios derived from NSL-KDD attack categories to assess the adaptive deception capabilities of the framework. Unlike purely simulation-based cybersecurity environments, the proposed system also collected live attacker interaction data during continuous multi-week deployment periods. This enabled the observation and analysis of genuine reconnaissance activity, malware propagation attempts, brute-force attacks, and post-compromise exploratory behavior under realistic operational conditions.

The reinforcement learning subsystem utilized protocol-specialized DQN agents implemented using PyTorch. Each service maintained an independent replay buffer containing 10,000 behavioral experiences, a target network updated every 200 optimization steps, and epsilon-greedy adaptive exploration policies with progressive decay scheduling. Initial policy bootstrapping was performed using approximately 3000 offline training episodes derived from NSL-KDD behavioral interaction patterns prior to live deployment. Continuous online learning was subsequently performed using real attacker interaction logs collected during deployment.

such as `wget`, `curl`, and automated credential probing behavior.

The proposed framework was evaluated within a controlled live deployment environment consisting of protocol-specialized SSH, HTTP, FTP, and Telnet deception services deployed across isolated virtualized infrastructure. During the evaluation period, the framework processed more than 830 protocol interaction events originating from 4 unique source IP addresses across heterogeneous deception services. The deployment additionally captured 8 malware-oriented payload interaction attempts including wget-based retrieval activity, command-based payload execution attempts, and privilege-oriented behavioral interaction patterns.

The screenshot displays the 'IoT AI Honeypot Command Center' dashboard. At the top, it shows four key metrics: 'Total AI Interactions' (830), 'Unique Attackers' (4), 'Most Targeted Service' (HTTP), and 'Malware Payloads Caught' (8). Below these, there are three horizontal bars representing 'Payloads Overview': 'Live Attack Feed' (red), 'Captured Malware' (green), and 'Payloads Overview' (blue). The main section contains two charts: 'Top Attacker IPs' (a table) and 'Targeted Services' (a bar chart). The 'Top Attacker IPs' table lists four IP addresses and their total interactions. The 'Targeted Services' bar chart shows the frequency of attacks on various services, with HTTP being the most targeted. To the right, a partial view of the 'AI Actions Deployed' bar chart is visible.

Attacker IP	Total Interactions
172.18.0.1	650
10.0.2.2	250
Unknown	11
10.0.2.15	6

Service	Frequency
FTP	150
HTTP	800
SSH	150
TELNET	200

Protocol-wise interaction activity demonstrated varying behavioral characteristics across services. HTTP-oriented services generated the highest interaction volume due to repeated reconnaissance-oriented probing and automated request behavior, while SSH and Telnet environments produced deeper command-oriented interaction persistence and privilege escalation activity. FTP interactions primarily involved firmware-oriented exploration and file-transfer-oriented reconnaissance behavior.

Unlike conventional cybersecurity systems that primarily evaluate detection accuracy or classification performance, the

proposed framework focuses on adaptive attacker engagement and behavioral intelligence collection. Consequently, evaluation was performed using deception-oriented behavioral metrics specifically designed to measure interaction realism, engagement persistence, deception effectiveness, and post-compromise behavioral continuity.

1) *Behavioral Engagement Index (BEI)*: The Behavioral Engagement Index (BEI) measures overall attacker interaction richness by combining session duration, command depth, privilege escalation activity, and honeypot interaction into a unified behavioral engagement score:

$$BEI = w_1D + w_2C + w_3P + w_4H \quad (4)$$

where D represents normalized session duration, C denotes normalized command interaction depth, P corresponds to privilege-oriented interaction activity, and H represents honeypot interaction frequency. Empirically selected weighting coefficients were assigned as $w_1 = 0.4$, $w_2 = 0.3$, $w_3 = 0.2$, and $w_4 = 0.1$ to prioritize sustained interaction persistence and behavioral exploration depth during adaptive deception evaluation.

Higher BEI values indicate stronger attacker immersion, deeper filesystem exploration, sustained post-compromise interaction, and increased operational trust in the deception environment.

2) *Deception Retention Rate (DRR)*: The Deception Retention Rate (DRR) measures the percentage of attacker sessions that continue meaningful interaction following adaptive deception exposure including staged credential disclosure, delayed authentication handling, privilege escalation interaction, or honeypot exposure:

$$DRR = \frac{\text{Retained Sessions}}{\text{Total Deception-Exposed Sessions}} \times 100 \quad (5)$$

Unlike simplistic session-duration metrics, DRR evaluates whether attackers remain behaviorally engaged after deception escalation rather than immediately disengaging from the environment.

3) *Escalation Persistence Score (EPS)*: The Escalation Persistence Score (EPS) measures attacker persistence following privilege-oriented interaction such as sudo attempts, root-level exploration, or administrative credential access:

$$EPS = \frac{\text{Commands After First Privilege Attempt}}{\text{Total Session Commands}} \quad (6)$$

Higher EPS values indicate sustained attacker interaction even after privilege-related behavioral stages, demonstrating stronger deception realism and post-compromise engagement continuity.

D. Protocol-Wise Behavioral Analysis

Experimental observations revealed substantial behavioral differences across protocol services. SSH environments achieved the highest observed behavioral engagement and

sustained privilege-oriented interaction persistence among all evaluated protocol services. because the environment provided unrestricted shell interaction, realistic filesystem exploration, operational shell history artifacts, staged privilege escalation opportunities, and progressive credential exposure. Attackers interacting with SSH environments frequently transitioned from reconnaissance-oriented behavior into sustained exploratory interaction, indicating strong perceived compromise realism.

Telnet services demonstrated the highest privilege persistence behavior as reflected by elevated EPS values. The BusyBox-oriented router simulation, Mirai-oriented malware bait interaction, NVRAM exposure, and wget/curl deception mechanisms successfully encouraged malware deployment behavior and sustained post-compromise interaction. Several Telnet sessions exhibited repeated payload retrieval attempts and recursive botnet-style interaction sequences characteristic of Mirai-like propagation activity.

HTTP interaction behavior was comparatively shallow because most observed activity consisted primarily of automated reconnaissance traffic, vulnerability scanning, and one-shot exploit validation attempts. Although the HTTP deception environment exposed SCADA-oriented dashboards, hidden credentials, and firmware upload functionality, many Internet-originated HTTP interactions remained short-lived due to the prevalence of automated scanning infrastructure.

FTP services demonstrated moderate interaction depth with several sessions involving malware-oriented file traversal, firmware exploration, and credential-oriented directory inspection. Uploaded payloads and transferred artifacts were successfully captured and logged for behavioral analysis.

Overall, the evaluation demonstrated that protocol realism strongly influenced attacker persistence and behavioral interaction depth. Interactive shell-oriented environments produced significantly deeper engagement compared to request-oriented services dominated by automated scanning traffic.

E. Adaptive Deception Behavior Analysis

Table III presents a comparative overview of representative IoT honeypot frameworks with respect to protocol coverage, reinforcement learning integration, orchestration capability, and deployment characteristics.

TABLE III: Comparison with Existing IoT Honeypot Frameworks

Framework	Services	RL Method	Orchestration Deployment	
ThingPot [2]	HTTP, XMPP	Static Logic	IoT Messaging	Real
U-PoT [1]	UPnP	Signature-Based	Device Emulation	Real
AIIPot [8]	HTTP	Adaptive RL	Local Interaction	Partial
Proposed	SSH, HTTP, FTP, Telnet	DQN-Based	MQTT-Driven	Real

As shown in Table III, most existing IoT honeypot systems primarily focus on single-service interaction environments or static deception architectures. In contrast, the proposed framework combines protocol-specialized DQN-driven adaptive interaction with MQTT-based distributed orchestration across heterogeneous IoT services, enabling scalable behavioral deception and sustained post-compromise interaction handling.

The reinforcement learning subsystem dynamically selected adaptive deception strategies according to evolving attacker behavior and protocol-specific interaction context. Among all adaptive actions, tarpit-oriented delayed-response strategies were selected most frequently because they effectively slowed aggressive reconnaissance activity while increasing interaction dwell time. Attackers exposed to adaptive delayed responses frequently retried commands, repeated interaction attempts, and continued exploratory behavior despite increased response latency.

Progressive honeypot exposure additionally proved highly effective for sustaining deeper exploratory interaction. Attackers encountering exposed credentials, cloud configuration artifacts, MQTT settings, and operational filesystem traces frequently initiated repeated traversal behavior, privilege escalation attempts, credential-oriented exploration, and recursive directory inspection.

Experimental observations further demonstrated that progressive trust calibration significantly influenced attacker persistence. Sessions involving delayed authentication handling and staged privilege escalation behavior often transitioned from shallow probing activity into extended post-compromise exploration. These observations suggest that gradual operational realism escalation improves attacker trust and reduces early honeypot suspicion.

Compared to static baseline interaction policies, the adaptive RL-driven framework consistently produced longer session duration, increased command complexity, higher behavioral engagement scores, improved deception retention, and more extensive malware-oriented interaction behavior. Static interaction environments frequently resulted in rapid disengagement following repetitive responses or unrealistic operational behavior.

Fig. 6 presents normalized behavioral deception evaluation metrics obtained during live deployment. The observed Behavioral Engagement Index (BEI) indicates moderate but meaningful attacker immersion and exploratory interaction behavior across protocol-specialized deception environments. Similarly, elevated Escalation Persistence Scores (EPS) demonstrate sustained attacker activity even after privilege-oriented interaction stages such as sudo attempts and root-level exploration. Although the Deception Retention Rate (DRR) remained comparatively lower, this behavior reflects realistic Internet-facing deployment conditions dominated by shallow automated reconnaissance traffic and short-lived botnet interaction patterns.

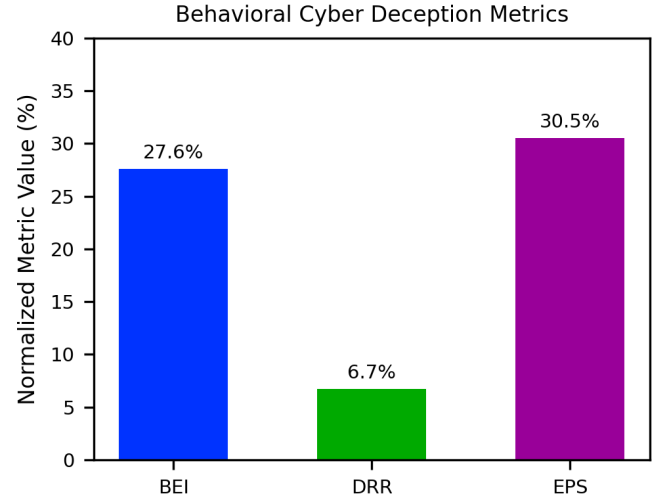


Fig. 6: Behavioral cyber deception evaluation metrics measured during live Internet-facing deployment. The framework achieved meaningful attacker engagement and privilege-oriented persistence while maintaining adaptive deception continuity across heterogeneous IoT services.

V. CONCLUSION

This paper presented an adaptive reinforcement learning-driven IoT honeypot framework designed to improve attacker engagement, deception realism, and behavioral intelligence collection within heterogeneous IoT environments. Unlike conventional static honeypot systems that rely on fixed interaction behavior, the proposed framework dynamically orchestrates deception strategies according to evolving attacker activity using protocol-specialized Deep Q-Network (DQN) agents integrated across SSH, HTTP, FTP, and Telnet services.

The framework introduced protocol-aware behavioral state modeling, progressive realism escalation, staged trust calibration, adaptive privilege-oriented interaction handling, and deception-driven operational artifact exposure to sustain post-compromise attacker interaction. Experimental evaluation within a controlled deployment environment incorporating structured simulation scenarios derived from NSL-KDD attack categories demonstrated that reinforcement learning-assisted adaptive deception significantly improved behavioral engagement depth, exploratory persistence, malware-oriented interaction activity, and deception continuity compared to conventional static or rule-based honeypot approaches.

Behavioral evaluation metrics including the Behavioral Engagement Index (BEI), Deception Retention Rate (DRR), and Escalation Persistence Score (EPS) further demonstrated that adaptive deception strategies can transform shallow reconnaissance-oriented activity into sustained post-compromise behavioral interaction. In particular, protocol-specialized shell-oriented environments such as SSH and Telnet produced substantially deeper attacker engagement due to operational realism, staged privilege escalation behavior, and

progressive credential-oriented interaction design.

Despite these contributions, several limitations remain. The current framework primarily focuses on SSH, HTTP, FTP, and Telnet environments, while modern IoT ecosystems increasingly rely on lightweight protocols such as MQTT, CoAP, Modbus, RTSP, and other industrial communication mechanisms. Additionally, although the framework performs adaptive action selection using reinforcement learning, interaction responses remain primarily template-oriented and do not yet incorporate semantic reasoning or human-like contextual interaction capabilities.

Future work will therefore focus on integrating LLM-driven semantic deception mechanisms capable of generating context-aware adaptive responses, autonomous dialogue-oriented interaction, and realistic command-level behavioral simulation. Additional future directions include attacker profiling and behavioral classification for distinguishing botnet scanners, automated exploitation frameworks, human operators, and malware-oriented adversaries. Expanding the framework toward distributed collaborative deception architectures and large-scale IoT protocol coverage will further improve behavioral realism, adaptive intelligence collection, and anti-fingerprinting capability in next-generation cyber deception systems.

Overall, the proposed framework demonstrates that adaptive reinforcement learning-driven deception orchestration can significantly enhance behavioral realism, attacker persistence, and post-compromise intelligence collection within controlled multi-service IoT deception environments, representing a promising direction for future intelligent cyber deception research.

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